

CHAMOUCOUANE, QC, Canada

WMO: 715230

Lat: 49.278N

Lon: 73.356W

Elev: 997

StdP: 14.17

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-35.1	-29.5	-41.5	0.5	-34.9	-35.3	0.8	-29.5	9.1	14.8	7.7	12.2	1.2	10	N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	23.4	83.8	67.4	80.1	65.5	76.6	64.2	71.0	77.5	68.9	75.8	66.9	73.5	4.5	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
68.9	110.4	73.7	66.7	101.9	71.7	64.6	94.5	69.3	35.6	77.9	33.7	75.9	32.1	73.8	77.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 9.1	7.7	6.9	-42.9	90.5	5.2	3.1	-46.7	92.7	-49.7	94.6	-52.7	96.3	-56.5	98.6		
(5)			-42.8	74.5	5.2	2.3	-46.5	76.1	-49.6	77.4	-52.6	78.7	-56.4	80.4		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	33.5	-0.5	2.1	15.2	31.5	46.8	57.5	62.0	59.8	51.4	39.3	25.3	9.3
	DBStd	24.22	14.94	14.04	14.35	9.57	8.50	7.83	5.61	5.93	7.80	7.55	10.02	13.28
	HDD50	7150	1566	1342	1078	554	168	20	0	3	74	343	742	1260
	HDD65	11588	2031	1762	1543	1004	567	247	127	180	413	797	1192	1725
	CDD50	1124	0	0	0	0	68	246	373	308	117	12	0	0
	CDD65	87	0	0	0	0	2	24	35	20	6	0	0	0
	CDH74	1341	0	0	0	0	112	443	453	268	64	1	0	0
	CDH80	328	0	0	0	0	24	133	111	48	12	0	0	0
	WSAvg	2.5	2.2	2.5	2.9	3.0	2.9	2.5	2.3	2.3	2.1	2.6	2.6	2.4
	PrecAvg	38.9	2.4	2.0	2.2	2.8	2.9	3.8	4.6	3.9	4.4	3.7	3.1	3.0
(15) Precipitation	PrecMax	45.6	4.9	4.0	4.2	4.5	5.1	5.9	7.6	6.1	8.1	5.6	5.0	6.7
	PrecMin	32.5	0.9	1.0	1.1	1.1	0.9	1.9	2.1	2.0	2.5	1.3	1.3	1.5
	PrecStd	3.2	0.9	0.7	0.8	0.8	0.9	1.0	1.2	1.0	1.3	1.1	0.9	1.1
	PrecStd	3.2	0.9	0.7	0.8	0.8	0.9	1.0	1.2	1.0	1.3	1.1	0.9	1.1
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	37.1	39.0	53.0	63.0	82.9	89.4	88.2	84.6	81.3	68.2	51.0	36.8
		MCWB	36.9	35.9	42.3	47.7	61.7	69.3	69.3	68.8	69.6	59.5	48.4	35.9
	2%	DB	32.7	33.1	45.2	57.4	76.7	83.7	83.0	80.6	74.1	61.2	45.9	33.5
		MCWB	32.5	31.4	37.7	44.9	58.5	66.2	67.7	67.8	66.0	55.2	44.0	33.2
	5%	DB	28.1	29.5	39.8	51.5	70.6	79.2	79.1	76.7	69.4	55.4	40.8	31.4
		MCWB	27.4	28.1	34.4	41.8	56.1	63.4	65.8	66.0	63.1	51.6	39.6	31.1
	10%	DB	23.0	25.6	35.4	46.4	64.5	74.3	75.3	72.9	65.1	51.1	37.1	28.5
		MCWB	22.3	24.3	32.2	38.9	53.2	61.3	64.5	64.1	60.2	48.4	36.2	27.9
	0.4%	WB	36.2	36.6	44.1	50.4	65.5	73.0	73.6	72.8	71.7	61.5	49.7	35.8
		MCDB	36.3	37.7	50.2	58.8	77.2	82.9	81.1	79.2	79.2	67.4	50.4	36.3
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	32.3	31.7	38.6	46.0	61.2	69.2	71.0	70.0	67.8	56.9	44.5	33.1
		MCDB	32.3	32.9	43.9	55.1	70.9	78.8	76.3	76.6	72.2	60.5	45.5	33.3
	5%	WB	27.0	28.0	35.3	42.9	57.8	66.5	68.8	67.9	64.4	52.3	39.6	30.7
		MCDB	27.4	29.2	38.8	50.4	67.8	74.4	75.5	74.4	68.4	54.6	40.7	30.9
	10%	WB	22.0	24.2	32.4	39.9	54.8	63.7	66.7	65.6	60.9	48.8	36.2	27.5
		MCDB	22.7	25.4	35.3	45.3	63.5	71.3	73.6	70.9	64.1	50.8	37.2	28.0
(35) Mean Daily Temperature Range	5% DB	MDBR	22.9	27.9	27.7	23.1	25.3	25.9	23.4	22.5	21.0	15.1	13.1	17.0
		MCDBR	20.4	21.2	30.2	31.6	37.3	34.0	30.4	28.8	27.7	24.7	16.9	13.4
		MCWBR	20.1	19.3	23.2	20.1	20.3	17.2	16.3	16.7	18.3	18.1	14.5	13.8
	5% WB	MCDBR	19.5	19.2	25.6	27.8	32.1	27.5	24.1	24.4	25.5	22.3	15.8	13.8
		MCWBR	19.5	17.6	19.7	18.8	18.6	15.9	14.6	15.5	18.2	17.4	14.2	13.8
(40) Clear-Sky Solar Irradiance	taub		0.265	0.281	0.290	0.342	0.344	0.369	0.386	0.368	0.356	0.320	0.320	0.263
	taud		2.237	2.212	2.247	2.312	2.447	2.405	2.379	2.431	2.451	2.529	2.323	2.291
	Ebn at Noon		250	274	292	287	290	282	275	276	268	261	225	235
	Edh at Noon		23	30	35	37	34	36	36	33	29	23	22	19
	RadAvg		414	749	1169	1485	1669	1776	1623	1354	1025	587	309	256
(45) All-Sky Solar Radiation	RadStd		52	92	115	120	151	132	148	87	88	58	36	34

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (10 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page